

# Software Quality Assurance Plan

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## Section 1: Purpose

The specific purpose of and scope of this particular SQAP is to cover the quality assurance of our Fundraiser software that we will be developing. This SQAP contains only 1 software item that will be covered which is our fundraising software and the intended use are for fundraiser hosts to keep track of everything on one seamless platform. The SQAP will cover the testing aspect of the software life cycle for the Fundraiser. We will be testing through implementation.

## Section 2: Reference Documents

- IEEE 730-2002

## Section 3: Management

### 3.1 Organization (copied)

Our project manager and team members who are partly independent in development will oversee V&V tasks. The leader can oversee testing and documentation review, while developers handle code verification. These are coordinated with configuration management and project management through regular meetings and shared platforms (Github).

### 3.2 Tasks

The SQAP applies to all phases of the project: requirements, design, development, testing and deployment. The main QA tasks include

- a) requirements and design reviews
  - i) Entry criteria: Draft requirements completed, and reviewed
  - ii) Exit criteria: Design approved for coding
- b) code reviews and testing
  - i) Entry criteria: Code is implemented
  - ii) Exit criteria: All tests are passed and the critical bugs are fixed
- c) configuration and quality audits before release
  - i) Entry criteria: all documents and test results are ready
  - ii) Exit criteria: QA approval for release

We will spend the first week on requirements and design review. Then, we will work for two weeks for Code Reviews and Testing. The following week will be the final audit and release approval

### 3.3 Roles and Responsibilities

The leader will be responsible for looking over and overseeing the requirements and reviews. For the developers, there should be

## Section 4: Documentation

### 4.2 Minimum Document Requirements

Software Requirements Description (TBD)

Software Design Description (TBD)

[Verification and Validation Plans](#)

[Software Project Management Plan](#)

[Software Quality Assurance Plan](#)

[Software Configuration Management Plan](#)

## Section 5: Standards, practices, conventions, and metrics

### 5.1 Purpose

We will be implementing unit tests as we develop to ensure quality. These quantifiable metrics include correct amounts of money being transferred, rounding accuracy, load-time, as well as the number of donations allowed in a set amount of time. A more rigorous set of numbers will be determined in the future once we implement said tests.

B. Conformance with these tests will be maintained and assured through repeated testing throughout the development phase.

### 5.2 Content

- a) Documentation standards: Our documentation will follow IEEE standards as listed under our reference documents as well as individual sections for each document.
- b) Design standards: We will have a uniform design process (Agile) and follow the suggested timeline.
- c) Coding standards: We will employ a uniform naming convention as well as alphabetical ordering of all methods and classes.
- d) Commentary standards: We will comment above every method with a brief description of its intended functionality.
- e) Testing standards and practices: We will have unit tests to ensure quality.
- f) Selected software quality assurance product and process metrics: Our metrics are quantifiable and in alignment with IEEE standards as listed in IEEE Std 982.1TM-1988 [B6] and IEEE Std 982.2TM-1988 [B7].

## Section 6: Software Reviews

### 6.1 Purpose

Software reviews should be done before code is merged. Software reviews can be done any time during the sprint. Software reviews should be accomplished by confirming that the code meets all standards, is of quality, and functions as intended. Should software not pass the review it should be modified to meet the software review.

### 6.2 Minimum Requirements

Software Specifications Review (SSR) : Verify that the requirements are complete and clear before the development

Architecture Design Review (ADR): Check that the website structure and flow meet the design plan.

Detailed Design Review (DDR) Confirm that each page and component is designed correctly

Verification and Validation Plan Review: Ensure the SVVP covers all the necessary tests.

Functional Audit: Test website features to confirm they work as intended.

Physical Audit: Check that all files and assets are included and organized.

#### ~~Managerial Reviews~~

Software Configuration Management Plan Review: Confirm Github and folder structures are correct.

#### ~~Post Implementation Review~~

## Section 7: Test

- Tests should ensure that the code being tested does not create errors in the product
- Tests shall ensure that the product works as intended
- Pull requests should be tested before accepting
- Testing should be composed of running the program with the code being tested

## Section 8: Problem Reporting and Corrective Action

When a problem is encountered during development it should be fixed immediately. During the testing and maintenance phases it should be put on a scrum board. A Scrum Board can be used to highlight and prioritize issues. A specific organizational responsibility regarding their implementation is to ensure it works.

## Section 9: Tools, Techniques, and Methodologies

- Tools: VS code for coding, GitHub for version control, Google chrome for testing, google docs and drive for documentation
- Techniques: Peer code review , sprint demos, manual testing of website features and functional audits

- Methodologies: Agile Scrum process with weekly sprints, continuous testing after each sprint, pull requests reviewed before merging and maintaining code and documentation standards according to IEEE guidelines.

## Section 10: Media Control

The software should not be distributed(as it is a website) but should still be tested every sprint. Additionally, this website should only be distributed as client side code.

## Section 16: SQAP Change Procedure and History

SQAP should be changed within this document and we can maintain a history of changes through google document's history feature.